

Supporting Information

Non-Invasive Point-of-Care Device to Diagnose Acute Mesenteric Ischemia

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MATERIALS AND METHODS

Gold Nanoparticle Synthesis

Gold nanoparticles (GNPs) were synthesized as previously reported^{s1}, in which trisodium citrate dihydrate (TSC) (Sigma-Aldrich, Oakville, ON, Canada) worked as both a reducing and dispersion stabilizing agent. The functionalization of the GNPs with DAnB was implemented as discussed in^{s2, s3}.

GNP Characterization

The optical properties of the GNPs were characterized using an Agilent 8453 UV-vis Spectrophotometer, while their surface charge at different stages of functionalization was

monitored using a Malvern Zetasizer Nano ZS. The diameter of the GNPs was estimated by measuring the diameter of 125 particles from transmission electron microscope (TEM) images that were collected using a Philips 410 TEM instrument, and calculating the average particle diameter. The impedance was measured at each stage of surface modification as well as at stages of adding the I-FABP and DAnB-GNPs using potentiostat/galvanostat SP-200 controlled by EC lab software from BioLogic Science Instruments Inc. (Knoxville, Tennessee). Moreover, the surface chemical bioengineering was confirmed using AFM imaging at each step of the modification using a Veeco di multimode V atomic force microscope (AFM).

GNP Functionalization with Detection Antibody

The maximum absorption of the plasmonic resonance is centered at 523 nm as seen in the UV-Vis absorption spectra of the GNP dispersion (the red curve of Figure S1(a)). The diameters of the GNPs were estimated by measuring the diameter of 125 nanoparticles from a TEM image (Figure S1(b)) and the estimates were assembled into a histogram of the size distribution (Figure S1(c)). The diameter of the GNPs was calculated to be 15 ± 2.5 nm. To monitor the surface functionalization of the GNPs, we followed the shift in the maximum plasmonic resonance absorption and the alteration in the ζ potential across the modification steps. The maximum of the GNPs' plasmonic absorption red shifted from 523 nm to 528 nm and 536 nm after coating with Cysteamine Hydrochloride and the DAnB, respectively (Figure S1(a)). It is agreed that the maximum absorption of the plasmonic resonance is sensitive to the surrounding medium of the GNPs^{s4, s5}. As the functionalization process of the GNPs with DAnB progresses, the nature of the GNPs' surrounding changes due to the changing of the dielectric constant, and consequently the position of the maximum plasmonic resonance, also changes. The functionalization of the GNPs was further confirmed by monitoring the variation in the ζ potential. Figure S1(d) shows the values of the ζ potential, where the as-prepared GNPs start with a ζ potential of -23 mV. By coating the GNPs with Brij molecules, it shifts to -15 mV. While the anchoring of the Cysteamine Hydrochloride dramatically moved the ζ potential to +10 mV, the anchoring of the DAnB retrieved it back to -16 mV. All of these proved that the coating of the GNPs with DAnB was successful.

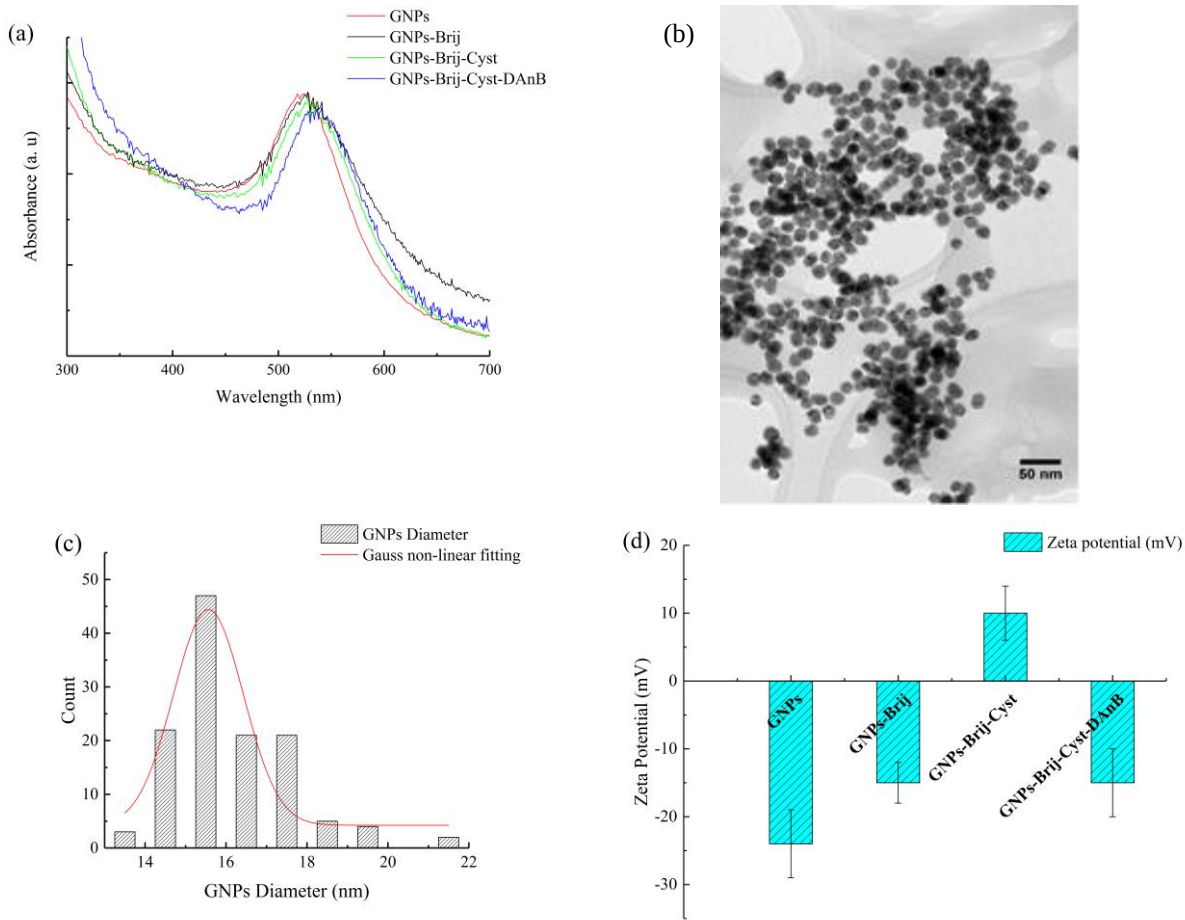


Figure S1: The evaluation of the GNPs surface functionalization. (a) the absorption spectra of the as-synthesized GNPs, after cysteamine hydrochloride and DAnB functionalization, (b) and (c) the transmission electron microscope image and the size distribution of the GNPs, respectively. The size distribution histogram showed that the GNPs have an average diameter of 15 ± 2 nm, where the dispersion state can be represented in the full width at half maximum (FWHM) of the Gaussian fitting of the histogram, and (d) the tracking of the ζ potential values of the GNPs at different stages of functionalization.

The size of the GNPs could also affect the capacitance signal and consequently the sensitivity of the IDE device. It can also influence the loading capacity of the detection antibody (DAnB) on the nanoparticles' surface, which adds more complexity to evaluate the role of the nanoparticle's size

in the sensing of the I-FABP. On the other hand, based on the simulation studies that we previously reported^{s6}, a three-fold changes in the size of GNPs induced only 4% change in the measured impedance magnitude. As a result, we kept the size of the GNPs (15±2 nm) fixed during the experiments.

IDEs Surface Functionalization

Functionalization of the silicon dioxide substrates of the IDEs was performed as follows (refer to Figure S2). Firstly, the IDEs were washed several times by sonicating in acetone and 2-propanol, respectively. After drying with N₂ gas, they were activated by oxygen plasma cleaning for 2 minutes and immersed in a 7% solution of (3-Aminopropyl) triethoxysilane (APTES) for 3 hours at 4 °C. Then, IDEs were washed three times both with ethanol and Milli-Q ultrapure water to discard the adsorbed and unreacted APTES. Thereafter, we incubated the IDEs in 5% glutaraldehyde solution overnight at 4 °C, washed afterward with Milli-Q water, and finally dried using N₂ gas. Before anchoring the CAnB, we treated the IDEs with 10 mM NaBH₄ for 20 minutes at 4 °C. We incubated the IDEs with different concentrations of CAnB (27, 54, and 108 µg/mL), where 30 µL of designated CAnB concentration solution was added to each electrode, and IDEs were kept at 4 °C overnight. Finally, the IDEs were washed with PBS buffer (10 mM) two times and Milli-Q water three times, was blocked with 1.0% Bovine serum albumin (BSA) solution for 1 hour, and washed thoroughly with PBS buffer and Milli-Q water.

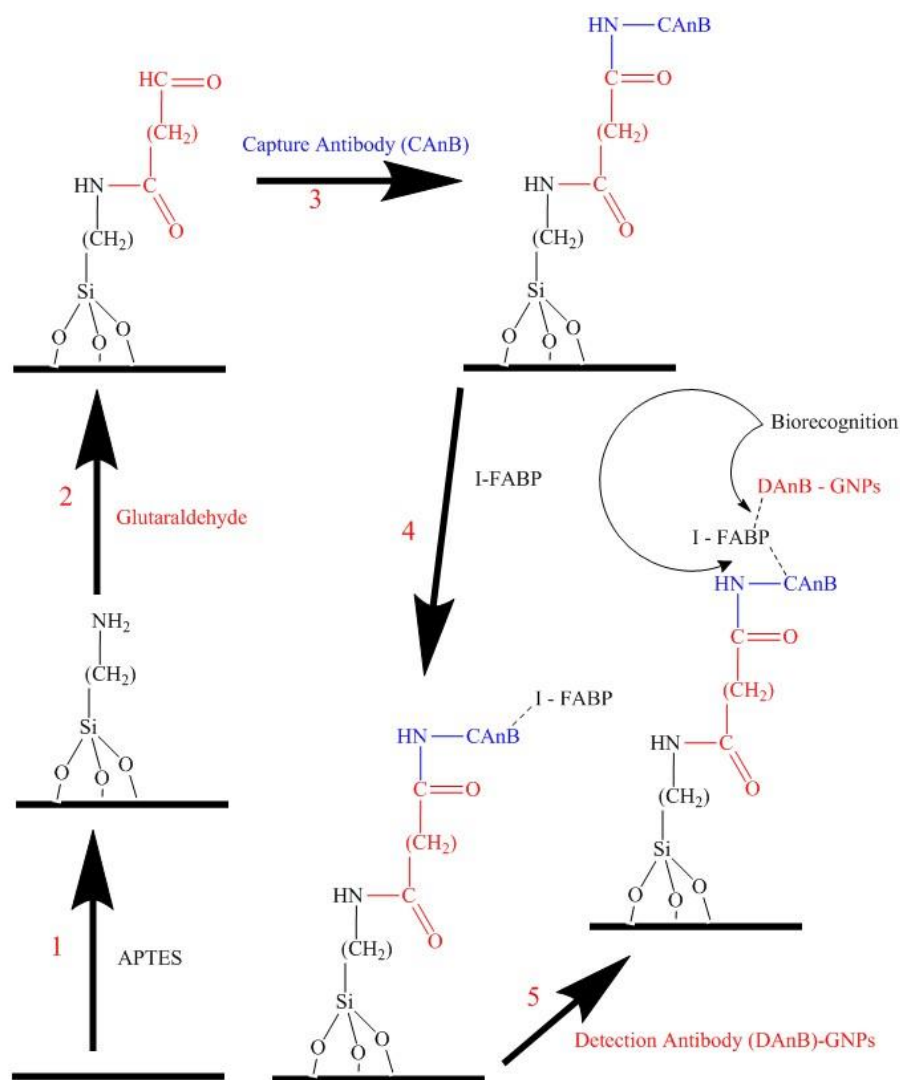


Figure S2: This figure is the schematic representation of the sensing steps of I-FABP on IDEs. By the sandwich assay. Steps 1 to 4 illustrated the preparation of the IDEs chip and binding of the CAnB. Step 5 is for detecting I-FABP using the DAnB-GNPs. Note that, in the negative control experiment, the I-FABP was replaced by H-FABP as a non-specific binding protein.

RESULTS AND DISCUSSION

IDE Surface Functionalization

Modifying the surface chemistry of the silicon dioxide gaps to accept the anchoring of CAnB required several stages. First, the silicon dioxide gaps between the IDEs chips were activated using oxygen plasma, creating hydroxyl functional groups that act as a foundation for the next

steps^{s7, s8}. Moreover, the generated hydroxyl groups by the oxygen plasma activation increased the hydrophilicity of the IDEs. Secondly, using APTES to modify the surface imparts amine groups to the SiO₂ surface, which can be utilized in the attachment of the CAnBs. The IDE chips were then immersed in glutaraldehyde solution, where the amine reacted with the aldehyde group forming an amide linker and providing a free aldehyde group for further functionalization^{s9, s10}.

The surface modification of the IDEs was monitored through following the variation in the IDEs surface topology using an AFM at each modification step. Also, the variation in the non-faradic impedance was observed to verify the progress in each step of the surface modification. Figure S3 depicts AFM images of the IDEs in the area between the gold fringes at different stages of surface coating. Before starting modification of the SiO₂, the surface of the IDEs was relatively uniform with a roughness of less than 2 nm. All non-chemically attached molecules were washed, and consequently, only the chemically attached molecules would remain on the top of the SiO₂ substrates. The roughness alteration across the surface modification increased from less than 2 nm to around 35 nm by the addition of the DAnB-GNPs at the last step of the detection, thus reflecting the chemical modification of the IDE surface.

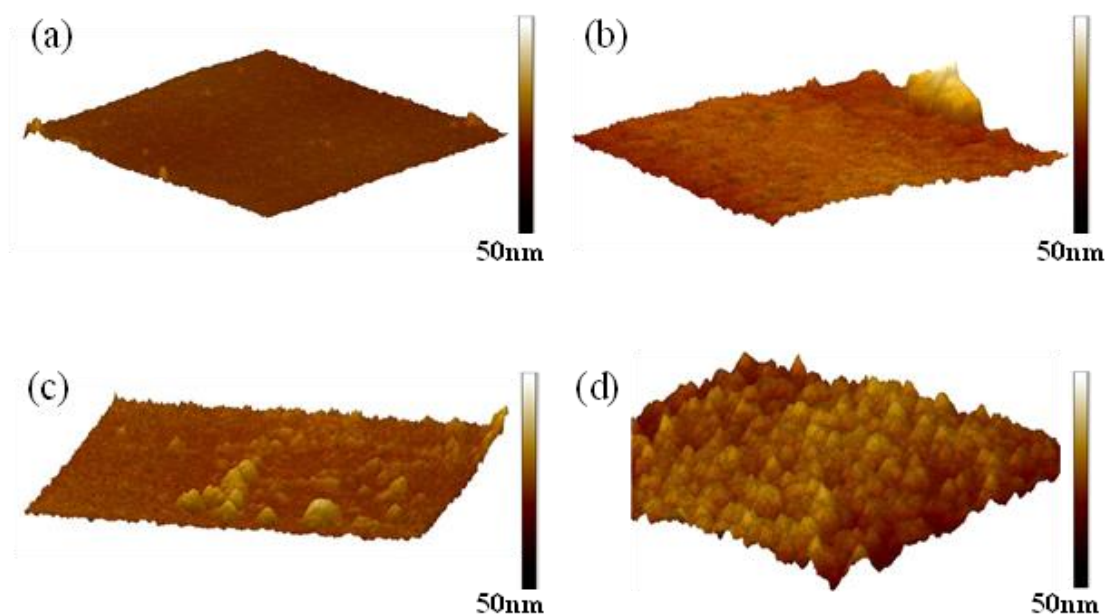


Figure S3: The AFM characterization of the electrode surface between the gold digits at different steps of the surface biomodification, where (a) is the bare surface before surface

modification, (b) is after anchoring of the capture antibody, (c) is after the addition of the I-FABP protein, and (d) is after the append of the GNPs coated with detection antibody.

Furthermore, we measured the impedance at each step of chemical surface modification and found that the magnitude altered by changing the surface groups. The variation in the impedance magnitude at the surface modification steps before adding the CAnB (data not shown) might be attributed to the alteration in the dielectric constant at the surface of the IDEs by changing the functional group. This change in the dielectric constant induced a capacitance behavioral alteration, which is seen in the variation of the impedance magnitude. The steps of the chemical surface modification of the SiO₂ intermittent substrate is schematically present in Figure S3. The binding of the CAnB to the aldehyde-functionalized IDEs was implemented through the amine-terminated group at the antibody. As shown in Figure S3, the surface modification process involved three steps to anchor CAnB to the silicon dioxide portion of the electrode, where the I-FABP detection took place in two steps. To avoid any possible contamination or deactivation, we performed all surface modification steps of the IDEs consecutively. The sensitivity and specificity of the IDEs to the I-FABP depended on the anchoring of the CAnB to their surface. However, the alteration in the impedance magnitude caused by the I-FABP alone was negligible. Figure S4 presents the impedance magnitude response of the IDEs at different stages of I-FABP sensing in a synthetic urine medium and the comparison with the impedance in the case of H-FABP as a negative control. The results of the surface functionalization steps of the GNPs are discussed in the supporting information.

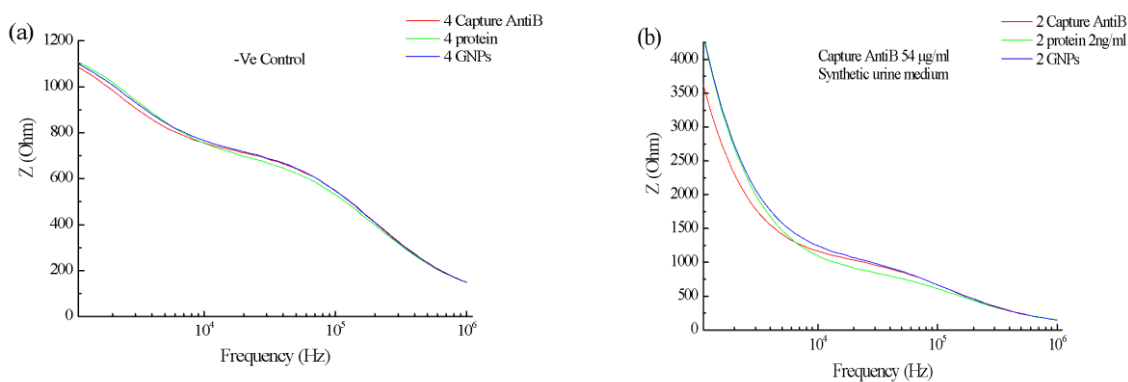


Figure S4: The impedance magnitude of IDEs measured at different stages of I-FABP sensing, where (a) is the negative control (H-FABP) reading and (b) is the measurements of 2 ng/mL of

I-FABP concentration. IDEs were functionalized with the CAnB of concentration 54 $\mu\text{g/mL}$. The target I-FABP protein was dissolved in synthetic urine medium. Figure (a) shows the change in the impedance magnitude by adding 2 ng/mL of I-FABP as compared with the negative control (H-FABP). By checking the scale of the Y-axis in both curves, it is easy to notice the difference in the impedance magnitude, where the maximum scale was 1200 Ohm in the case of the negative control (Figure a), while it was 4250 Ohm in the case of 2 ng/mL I-FABP (Figure b). We calculated the exact value of the impedance magnitude generated by the I-FABP as mentioned in the manuscript by subtracting the impedance magnitude of the I-FABP step from that of the GNPs-DAnB step (please refer to Figure S2).

The Role of GNPs in the Impedance Enhancement

The attachment of the I-FABP to the CAnB modified IDEs resulted in an insignificant change in the impedance magnitude, which necessitates a mechanism to amplify electrical impedance sensitivity. The proposed mechanism of the testing setup is shown in Figure S5(a), where GNPs were used to enhance the impedance signal by sandwiching the I-FABP with the anchored DAnB. Specifically, the presence of the GNPs bound to the substrate between the IDEs capturing the analyte I-FABP enhance the sensitivity of biosensor. This visible alteration allows for measuring impedance at a range of frequencies between 1 kHz to 1 MHz to detect the presence and concentration of the I-FABP. By measuring impedance over this short range of frequencies, much simpler (and miniaturized) electronics can be used. Additionally, by measuring impedance at only a small number of frequencies, together with the standard concentration of I-FABP, simple calibration curves can be used to determine the concentrations of I-FABP. There is no evidence of prior art based on preliminary intellectual property scans, research reports, or industry comments where a biosensor employs GNPs to enhance the change in impedance of bound I-FABP for AMI testing. Previously, we reported theoretically and practically^{s3}, the importance of the GNPs in boosting the impedance signal and consequently increasing the sensitivity of the IDE biosensor. Here we presented the utilization of the GNPs as an impedance signal booster based on its chemical functionalization capabilities to enhance the sensitivity of the IDEs biosensor to I-FABP by coating with DAnB. The circuit representing

the impedance of the IDEs is shown in Figure S5(b). The total impedance of this circuit is presented in Equation (S1).

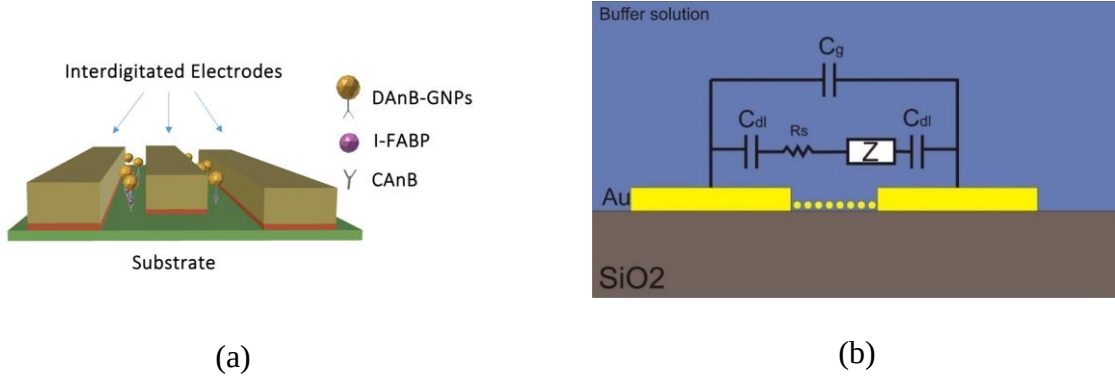


Figure S5: (a) The proposed impedance-based gold nanoparticle (GNPs) enhanced biosensor measurement method, where gold circles stand for GNPs, (b) equivalent model circuit of the IDEs impedance system.

$$Z_{net} = (R_s + Z + 2Z_{dl}) \parallel (1/j\omega C_{geo}) \parallel (Z_{par}), \quad (S1)$$

where Z_{net} is the net impedance of the system, R_s is the resistance of the solution, Z is the extra impedance introduced by the bound GNPs, Z_{dl} is the impedance of the double layer, C_{geo} is the dielectric (geometric) capacitance of the system, and Z_{par} is the parasitic impedance of the substrate. The values j and ω are the imaginary number and angular frequency, respectively. Z_{dl} is generated as a result of the formation of the ionic double layer at the charged surface. Under measurement conditions, a double layer of ions forms at the interface of any charged surface, which functions as a capacitor and can be represented as C_{dl} . This double layer behaves like a capacitor and contributes an impedance Z_{dl} to the system (one for each electrode finger in the pair). The value of Z_{dl} is presented in Equation (S2), where it can be noticed that the only variable parameter is the capacitance of the double layer in the case of using the same electrolyte for a given frequency.

$$Z_{dl} = 1/j\omega C_{dl}. \quad (S2)$$

When the DAnB-GNPs encounter an I-FABP attached to the IDEs surface by the CAnB, the detection antibody binds to that I-FABP molecule. This adds new charged surfaces to the IDEs in the gap between adjacent gold fingers and subsequently to the circuit. This new charged

surface element contributes to capacitance in the net impedance of the circuit (represented by Z in equation 1). As the number of anchored GNPs increases due to the presence of target I-FABP molecules, the number of capacitors that are created from the GNPs' charged surfaces (C_{GNP}) increases. The total capacitance ($C_{GNPTotal}$) from the GNPs is given by Equation (S3). As the number of GNPs increases, $C_{GNPTotal}$ decreases and therefore the resulting impedance (Z) increases, which affects the total impedance of the IDEs.

$$C_{GNPTotal} = [(1/ C_{GNP1}) + (1/C_{GNP2}) + (1/C_{GNP3}) +]^{-1} \quad (S3)$$

The sandwich technique that we used to detect the I-FABP is not new as it has been reported in literature. The design can be applied to different sensing applications, such as ELISA, optical and electrochemical sensing. However, in this article,

1. We used non-faradic impedance-based biosensor as shown in Figure S5. The usage of gold nanoparticles between the electrodes to amplify the non-faradic electrochemical impedance signal to boost the sensitivity of the biosensors is novel.
2. We applied the sandwich assay using gold nanoparticles functionalized with DAnB to detect I-FABP. It is the first time reported to detect acute intestinal ischemia using such a method, which is the novel.

In the case of the negative control (H-FABP), the impedance magnitude did not show any significant change after the addition of the DAnB-GNPs as shown in Figure S4(a). This is attributed to the lack of attachment of the DAnB-GNPs to the H-FABP molecules at the IDEs surface. This insignificant change in the impedance proves the high specificity of the IDEs toward I-FABP. Alternatively, when I-FABP is used, there is a significant change in the impedance magnitude after the addition of the DAnB-GNPs to IDEs, as shown in Figure S4(b). It was found that the addition of the DAnB-GNPs increased the impedance magnitude which might be attributed to the reduction of the total capacitance of the system. The functionalization of the IDEs with CAnB was found to be an important factor in determining the sensitivity of the sensor system to I-FABP. Three different concentrations of CAnB were used to coat the IDEs, and the sensing performance was measured.

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